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S2

Started on	Thursday, 1 October 2026, 9:28 AM
State	Finished
Completed on	Thursday, 1 October 2026, 9:29 AM
Time taken	47 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  // Function to check if a pair with the given difference exists
5  int findPairWithDifference(int* arr, int n, int k) {
6      int i = 0;
7      int j = 1;
8
9      // Search for the pair using two pointers
10     while (i < n && j < n) {
11         // Ensure i and j are distinct indices
12         if (i == j) {
13             j++;
14             continue;
15         }
16
17         int diff = arr[j] - arr[i];
18
19         if (diff == k) {
20             return 1; // Pair found
21         }
22         else if (diff < k) {
23             j++; // Increase the difference by moving the right pointer
24         }
25         else {
26             i++; // Decrease the difference by moving the left pointer
27         }
28     }
29
30     return 0; // No such pair exists
31 }
32
33 int main() {
34     int n, k;
35
36     // Read the number of elements
37     if (scanf("%d", &n) != 1) {
38         return 0;
39     }
40
41     // Allocate memory for the array
42     int* arr = (int*)malloc(n * sizeof(int));
43     if (arr == NULL) {
44         return 0;
45     }
46
47     // Read the array elements
48     for (int i = 0; i < n; i++) {

```

```
49 |         if (scanf("%d", &arr[i]) != 1) {  
50 |             free(arr);  
51 |             return 0;  
52 |         }
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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